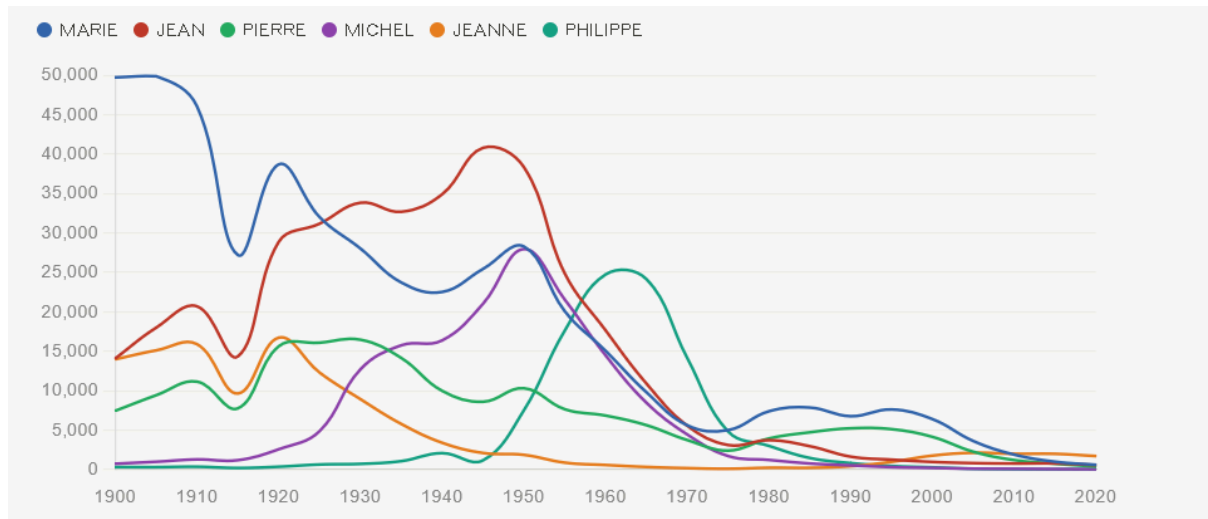


Week 1 : Curves idea for the different visualisations

Visualisation 1 : Temporal evolution :

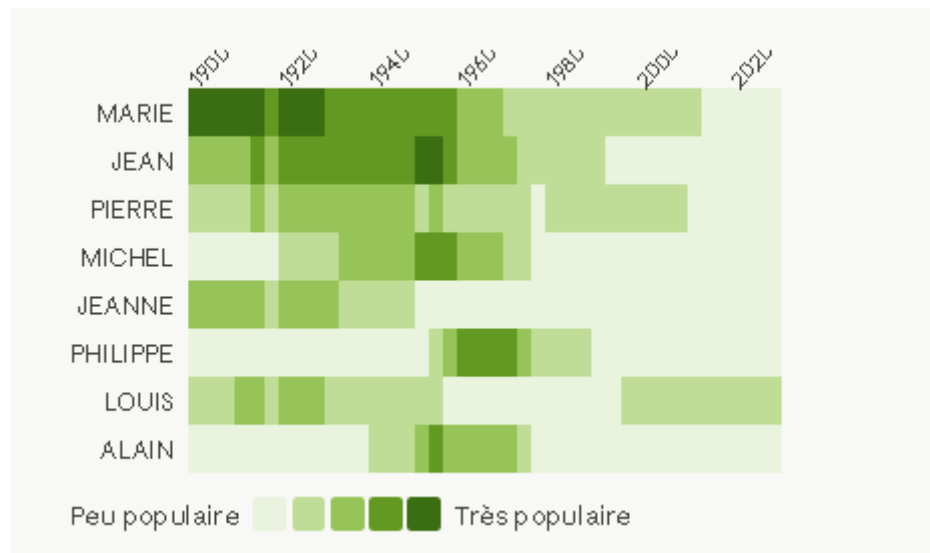
Idea 1 : Line chart



Pros: Very readable for comparing a few names side by side; peaks and drops are immediately visible, making it easy to spot sudden rises or falls; the x-axis naturally encodes time so the viewer always knows exactly which year they are looking at; works very well with an interactive search bar that lets users add or remove names on demand; precise values can be read directly from the y-axis.

Cons: Becomes a confusing "spaghetti chart" as soon as more than five or six names are displayed at once; hard to see global trends across all names since the eye focuses on individual curves; names with similar popularity levels produce overlapping lines that are nearly impossible to untangle even with different colors.

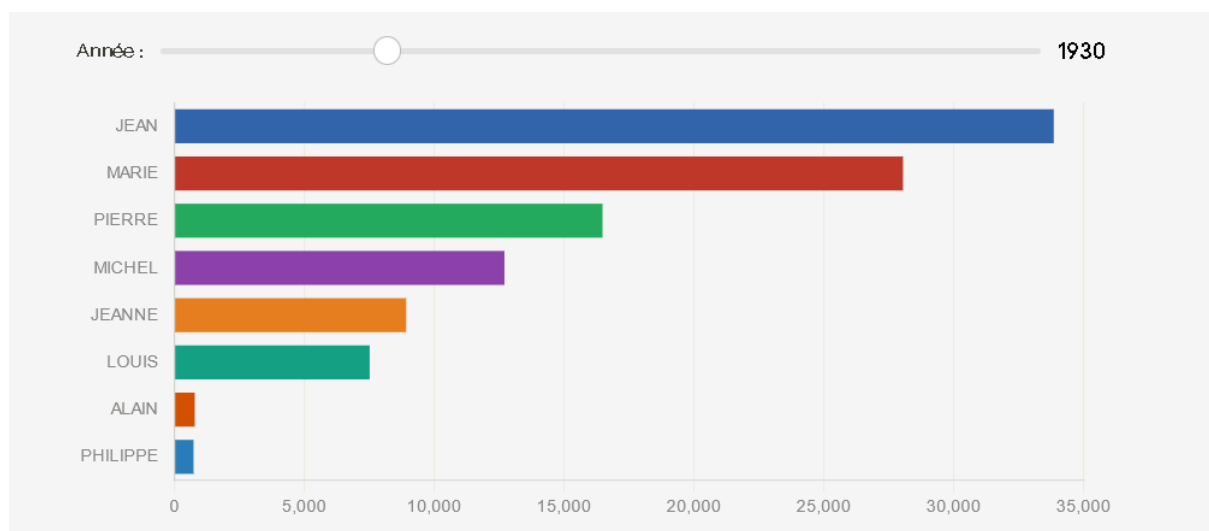
Idea 2 : Heatmap



Shows all names at a glance in one compact view without any interaction required; color makes broad patterns and eras of popularity very easy to spot instantly; great for comparing many names simultaneously and identifying which decades were dominated by which names; scales well to large datasets without becoming cluttered.

Cons: Hard to read precise values since the viewer can only estimate a range from the color; color perception varies significantly between people and those with color blindness may struggle to distinguish adjacent shades; names with moderate and stable popularity can look identical to one another; the ordering of rows is arbitrary unless sorted deliberately, which adds a design decision.

Idea 3 : Animated bar chart

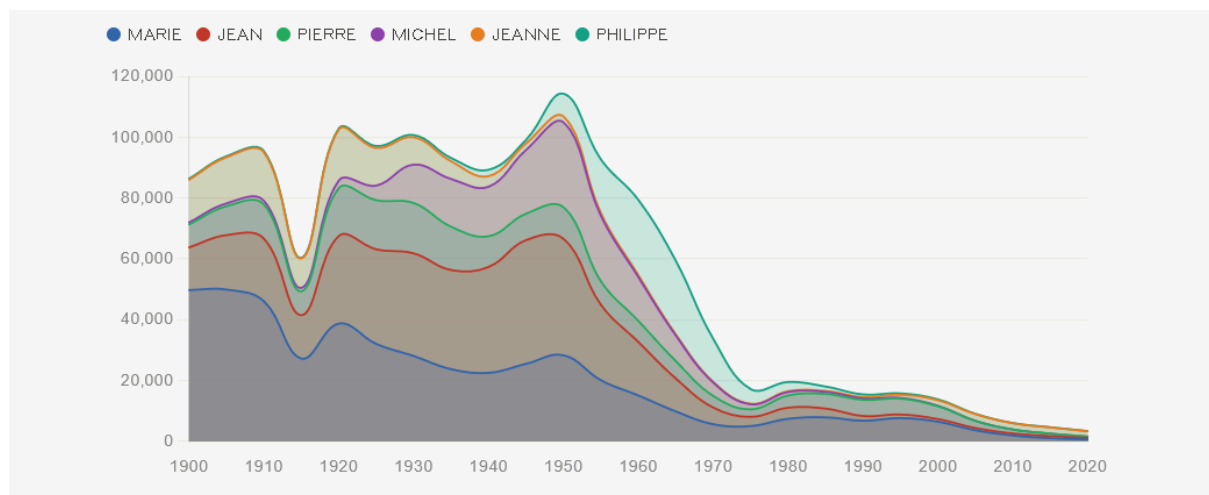


Pros: Visually engaging and dynamic, which makes it memorable and appealing for a general audience; spectacular rises and falls are obvious and create a narrative effect; easy

to understand intuitively even without any data knowledge; shows the competitive ranking between names in a compelling way.

Cons: Hard to analyze calmly since everything is moving and the viewer cannot pause to reflect; names that fall out of the top ranks disappear entirely from view, making it impossible to follow their full trajectory; not suitable for academic or rigorous analysis since comparisons across time require the viewer to remember previous frames; the animation can feel gimmicky if the changes between years are small.

Idea 4 : Stacked area chart

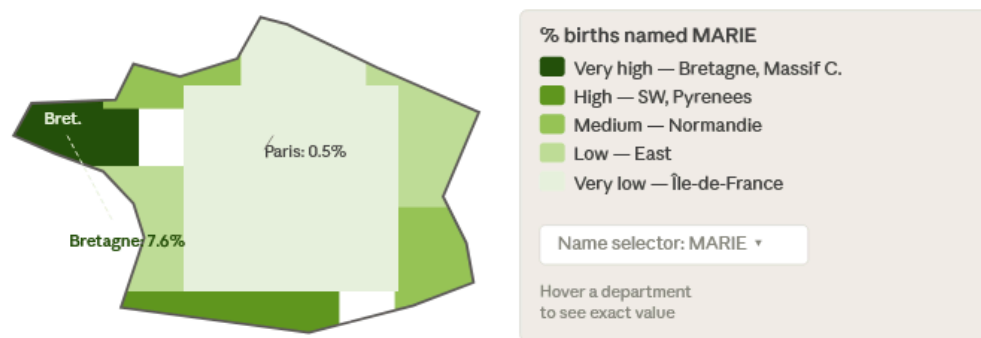


Pros: Beautiful and aesthetic, giving a strong visual impression; shows the overall volume of all names combined over time; good for seeing which era was globally dominated by a small set of names versus a more distributed landscape; conveys the idea of cumulative popularity naturally.

Cons: Very hard to read a single name precisely since only the bottom layer has a stable baseline; names placed in the middle or top of the stack are almost impossible to compare across years; can be deeply misleading if one name dominates the scale and compresses all others visually; adding or removing a name changes the shape of every other layer, making interactive filtering difficult.

Visualisation 2 : Regional effect:

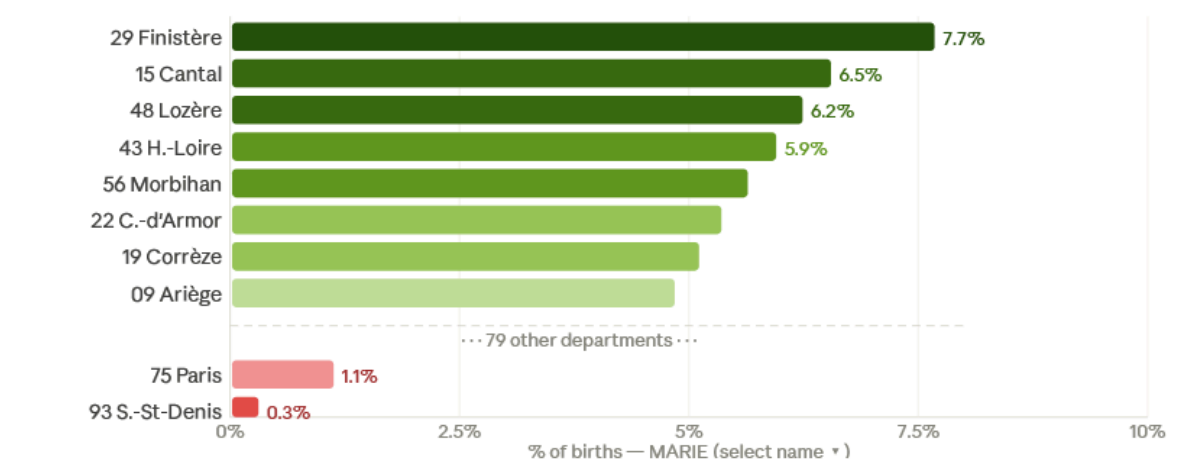
Idea 1 : Choropleth map:



Pros: Geographic patterns are immediately visible without any explanation needed; the format is completely intuitive since everyone knows how to read a map; spatial clusters such as Brittany or the Paris region stand out naturally; hover interactions can provide precise department-level values on demand.

Cons: It is very hard to compare multiple names at once since only one name can be mapped at a time; precise values are difficult to extract from color alone without hovering; small departments like those in the Paris region are visually underrepresented despite having large populations; color scales require careful calibration to avoid misleading the viewer.

Idea 2: Ranked horizontal bar chart

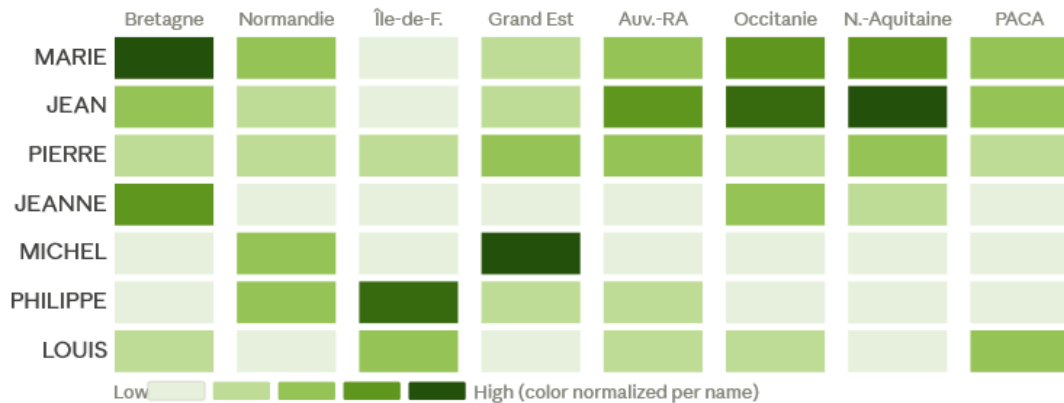


Pros: The ranking is crystal clear and exact percentages are directly readable from the bar lengths; extremely easy to spot extreme outliers at both ends of the spectrum; sorting the bars makes the comparison between departments straightforward and unambiguous; works well with a name selector to update the chart interactively.

Cons: Displaying all 95 departments creates a very long chart that requires scrolling and is cognitively overwhelming; there is no geographic context whatsoever so the viewer cannot

tell whether high-ranking departments are geographically clustered or scattered; showing only the top and bottom departments means the middle of the distribution is hidden.

Idea 3 : Heatmap - all names x all regions:

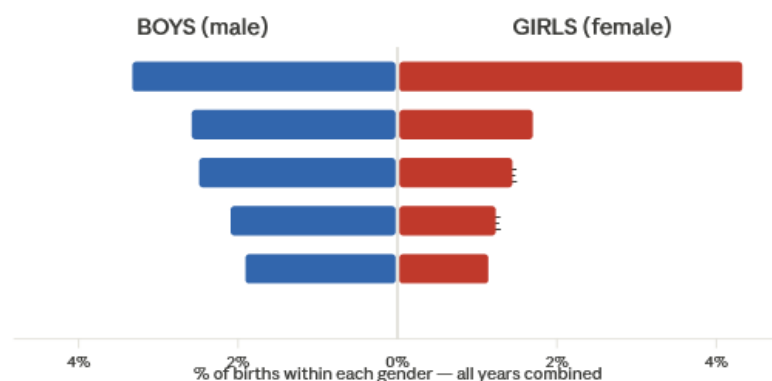


Pros: All names and all regions can be compared simultaneously in a single compact view without any interaction; makes it immediately obvious which names are uniform across France and which are concentrated in specific regions; a well-designed color scale lets the viewer scan for patterns across both dimensions at once.

Cons: Averaging values per region hides within-region disparities, for example a name could be very popular in one department of a region but absent in all others; the column order carries no geographic meaning so spatial clusters are not preserved; the matrix can become difficult to read if many names and many regions are included simultaneously. .

Visualisation 3 : Gender effect:

Idea 1 : Mirrored bar chart - top names per gender side by side

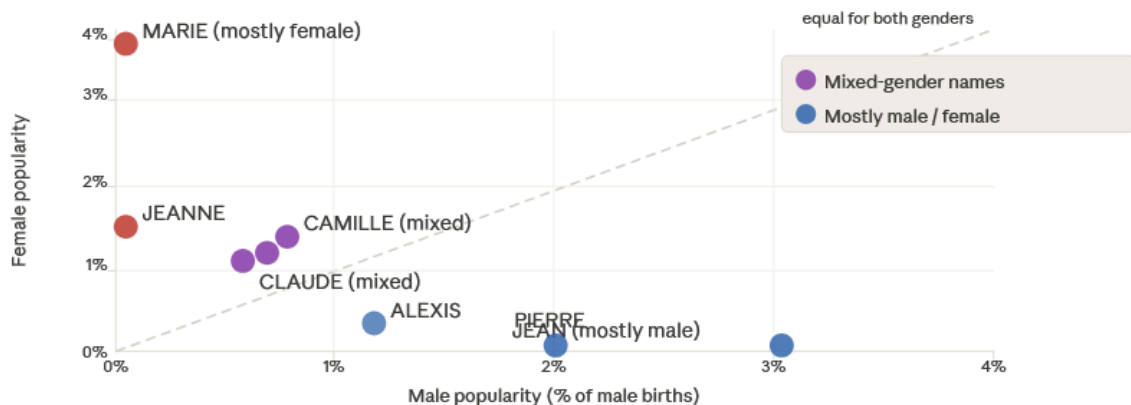


Pros: It immediately shows the most dominant names for each gender and makes the asymmetry between male and female naming patterns visible at a glance; the symmetric

layout creates a natural comparison between the two sides; the format is intuitive and requires no data literacy to understand.

Cons: Only the top names are shown so mixed-gender names that appear on both sides get completely lost; it gives no information about how popularity evolved over time; the chart is static and cannot answer questions about trends or gender shifts within a single name.

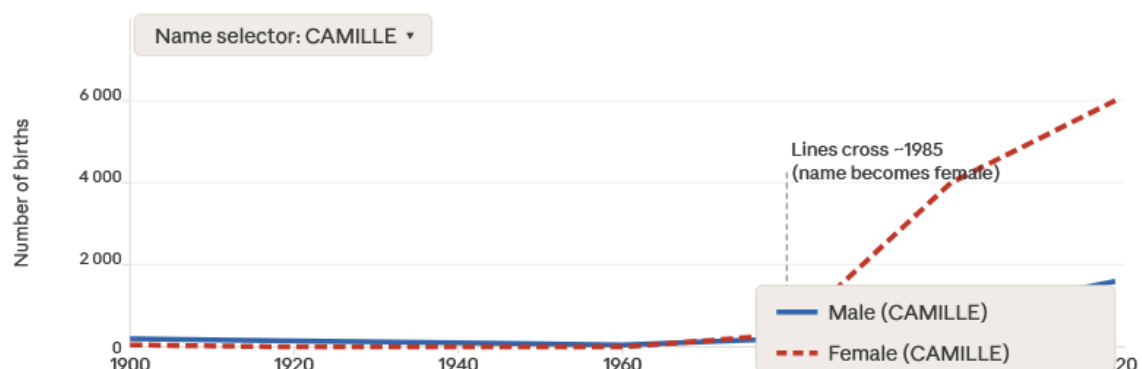
Idea 2 : Scatter plot - male popularity vs. female popularity per name



Pros: It directly reveals which names are gender-specific (near the axes) versus mixed-gender (near the diagonal), and shows the full spectrum in a single view.

Cons: Names tend to cluster near the axes making them hard to distinguish from one another, and there is no temporal information so you cannot see how the gender balance of a name changes over time.

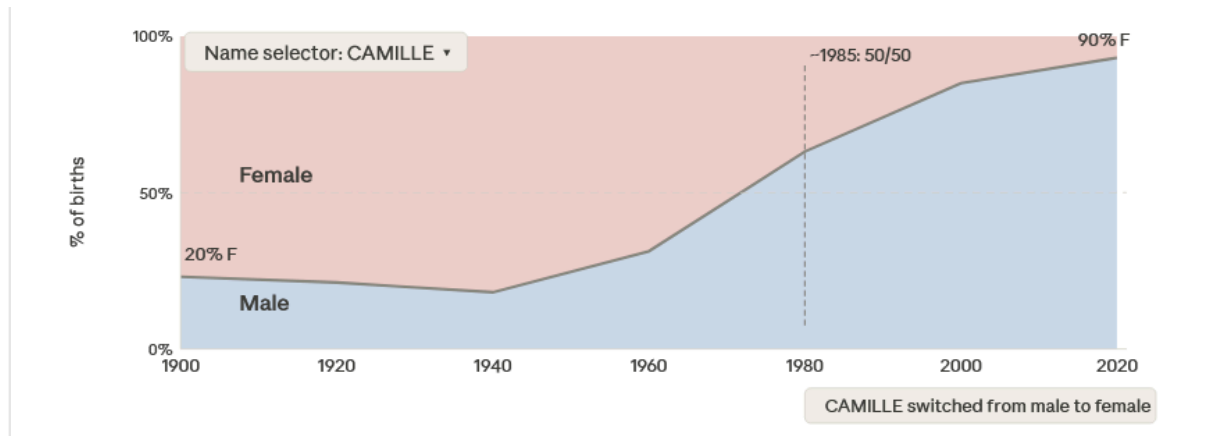
Idea 3 : Dual line chart - evolution of a mixed-gender name over time



Pros: It directly answers whether a mixed-gender name evolves consistently for both sexes, and makes trend crossover points - where a name switches from male-dominant to female-dominant, very easy to spot.

Cons: It only works well for names given to both sexes, and comparing several names at once quickly makes the chart very cluttered and hard to read.

Idea 4: Proportion chart - gender balance of a name over time



Pros: It dramatically shows how the gender balance of a name shifts over time, and makes it immediately obvious when a name has completely switched from one gender to the other.

Cons: Like the dual line chart, it only works meaningfully for mixed-gender names, and comparing multiple names at once requires either separate charts or additional interaction, which increases complexity considerably